

Question 5
Not yet answered
Marked out of 1.00

For each component in top-level entity:

Select one or more:

- ☐ a. When instantiated, mapping its ports is optional.
- ☒ b. When instantiated, it should have unique label.
- ☒ c. It can be instantiated multiple times.
- ☐ d. It can be declared multiple times.

Question 6
Not yet answered
Marked out of 1.00

The following VHDL statements deduce a flip-flop :

Select one or more:

- ☒ a. if (clk'event and clk='1') then q<= d;
- ☒ b. if (rising_edge(clk)) then q<= d;
- ☐ c. if (clk='1') then q<= d;
- ☐ d. q<= d when (clk='1') else q<= 0;

Question 7

Not yet answered

Marked out of
0.60



To design a counter that counts the even numbers from 0 to 20, how many D-flip flops we will need ?

Select one:

☐ a. 3

☐ b. 4

☒ c. 5

☐ d. 6

Quiz navigation

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25					

Finish attempt ...

Question 8

Not yet answered

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0.50



The ASSERT statement is used to check output values against expected values. select the Severity options of this command:

Select one or more:

☐ a. Note

☐ b. failure

☐ c. Wrong

☐ d. warning

☐ e. Error

☐ f. Correct



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Question 9

Not yet answered

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5.00

Design a simple ALU. There are 2 inputs, A and B, and each one is stored in 16-bit register and a 3-bit opcode. The output is a 17-bit register Y. The opcode controls ALU functions as follows:

Opcode	Operation	Function
000	$Y \leq A$	A
001	$Y \leq A + B$	Add
010	$Y \leq A - B$	Subtract
011	$Y \leq A \text{ AND } B$	AND
100	$Y \leq A \text{ OR } B$	OR
101	$Y \leq A + 1$	Increment A
110	$Y \leq A - 1$	Decrement A
111	$Y \leq B$	B

Write the full VHDL code for your design use your student ID as name of your top-level entity
Hint: use multiplexer and simple logic and arithmetic assignments.

Quiz navigation

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24				

Question 10

Answer saved

Marked out of
0.60



The V in VHDL stands for?

Select one:

- ☐ a. Vast
- ☐ b. Version 5
- ☐ c. Verilog
- ☒ d. Very High Speed IC
- ☐ e. Very

[Clear my choice](#)

[Previous page](#)



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Activate Windows
Go to Settings to activate Windows.

```
Q <= '0';  
else  
if (rising_edge(B)) then  
Q <= N;  
end if;  
end if;  
end process X;  
end arch;
```

Select one:

- ☐ a. 4-bit register with Synchronous reset
- ☐ b. 4-bit register with Synchronous reset and enable
- ☐ c. 4 D latch with enable
- ☐ d. 4-bit register with Asynchronous clear
- ☐ e. 4-bit register with Asynchronous clear and enable
- ☐ f. 4-bit register without clear, reset and enable

12
answered
out of

-- Consider the state table bellow, select the right choice to make the code bellow a
described by this state diagram?

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Calculators are not allowed

Note: Answer all questions

Q1: Say that $a(7:0) = "00110011"$ and $b(3:0) = "1111"$. Determine the values produced by the assignments below. (answer only 5) (5 marks)

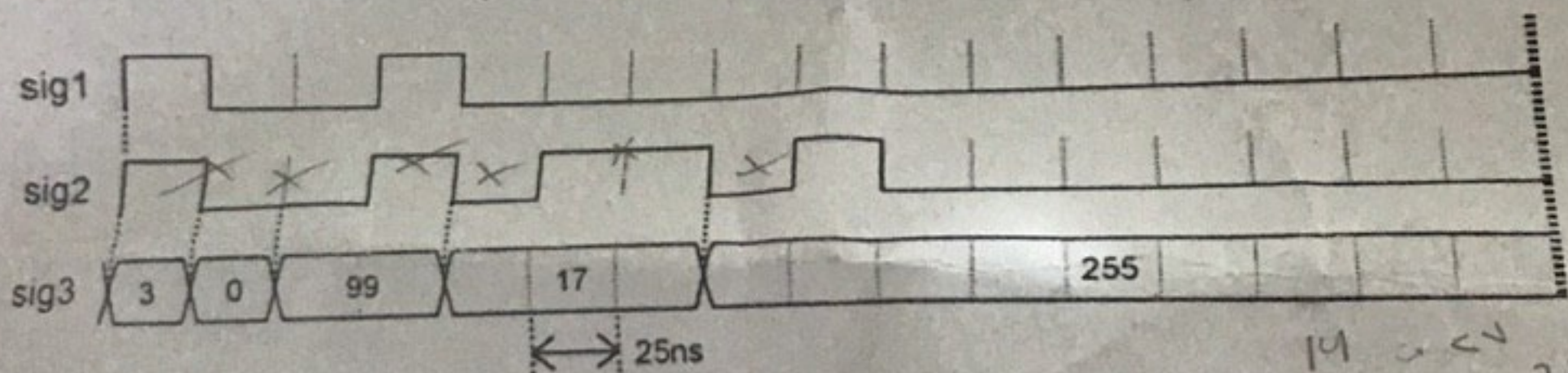
- $a(7 \text{ DOWNTO } 4) \text{ NAND } "0111"$
- $a(7 \text{ DOWNTO } 4) \text{ XOR NOT } b$
- $"1111" \text{ NOR } b$
- $b(2 \text{ DOWNTO } 0) \text{ XNOR } "101"$
- $a(2 \text{ DOWNTO } 0) \& "000"$
- $a(5) \& b(5) \& a(5 \text{ DOWNTO } 2)$

5432 1100

Q2: Implement a 4x1 multiplexer with: (10 marks)

- where inputs are single bit and using only logical operators (gate level design).
- N-bit inputs specify N using GENERIC and using WHEN statement.

Q3: Write the testbench code to generate the signals in the figure bellow: (10 marks)



Q4: Write the code for a zero to nine counter with reset signal input and the output (state value) is displayed using 7-segment displays HEX0. (15 marks)

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Good luck although you shouldn't need it!

Q4) (10 Marks)

- (a) Assuming architecture C in your lecture notes, determine the addressing mode and describe the following instructions as one of the following (a) Register-based, (b) Data transfer or (c) Control (ANSWER 5 ONLY)

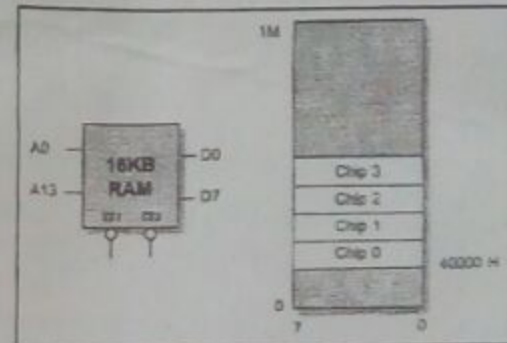
SUB R4,R2,R3 ; AND R1,R2,R3 ; ROR R1,R2,#3 ; STORO R1, R2,#1234 ; JMP #1234 ; BRZ R1,#1234

(5 Marks)

- (b) Assuming Architecture C in your lecture notes, write the instruction word for the instruction MOVI R3, #1234

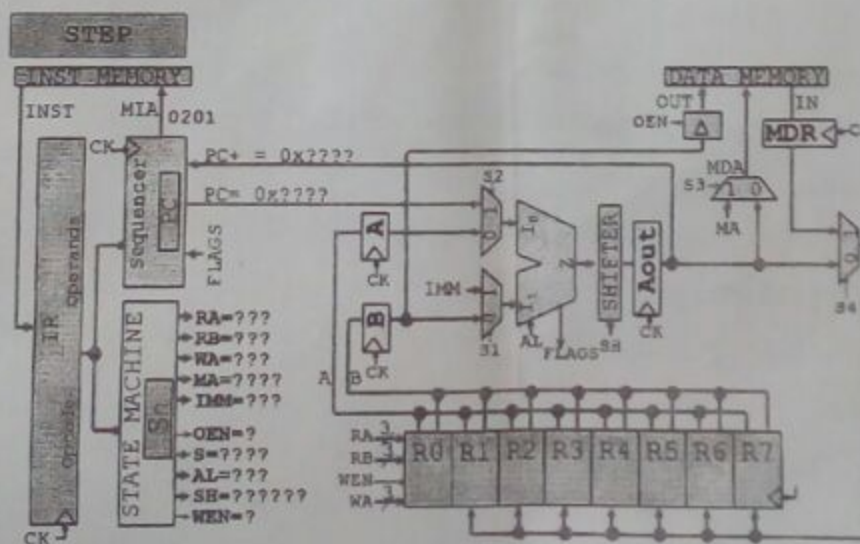
(5 Marks) (opcode for movi is 10000)

- (c) Show how to connect 4 of the following 16KB memory module starting from memory location 30000H onwards to produce 64KB memory capacity as shown in the memory map. Show the appropriate decoding logic for each module. Assume 20 bit address bus (A19-A0).



- Q5) Fill the following table with the control signals for each of the execution steps in architecture C (Multicycle) for the instruction BRNZ R3,0032 (10 Marks)

Step	PC=	PC+=	RA=	RB=	WA=	MA=	IMM=	OEN=	S[1-4]=	AL=	SH=	WEN=
Single Cycle	0x2222	0x2222	???	???	???	???	0x2222	?	???	???	???	?
Fetch	X	X								111		
Reg Read				011				0				0
Execute										101		
Mem Write	X	X	X	X	X	X	X	X	X	X	X	X
Reg Write	X	X	X	X	X	X	X	X	X	X	X	X



END OF QUESTIONS, GOOD LUCK!

- A. Design this circuit for 4 stages using VHDL (10 Marks)
- B. The entity shown in dashed box has the interfaces as follows:

```

ENTITY MUXED_FF IS
PORT (clk, D, S, Load : IN STD_LOGIC;
      Q : OUT STD_LOGIC);
END ENTITY;

```

Assume you have this entity in your library. Write VHDL to design the same circuit assuming the number of stages is a **GENERIC** named **N** and using **GENERATE** statement. (10 Marks)

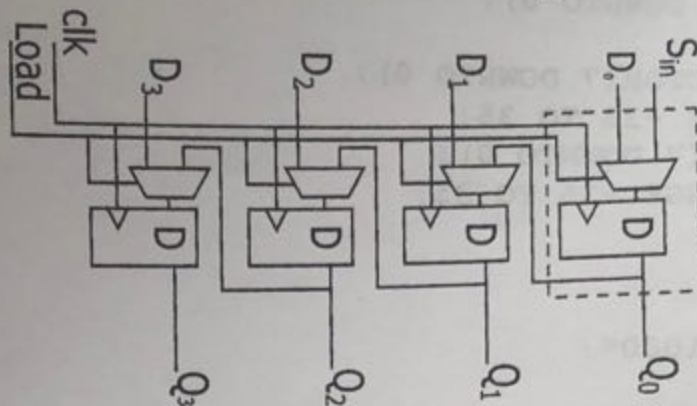


Figure Q3

Q4- Write a complete a testbench VHDL code of the processes that generate the signals A, B, C, D and E of figure Q4. Employ **WAIT FOR** statement. Each stimuli should be generated in a separate process. Note that A and B are periodic signals and should be generated indefinitely (10 Marks)

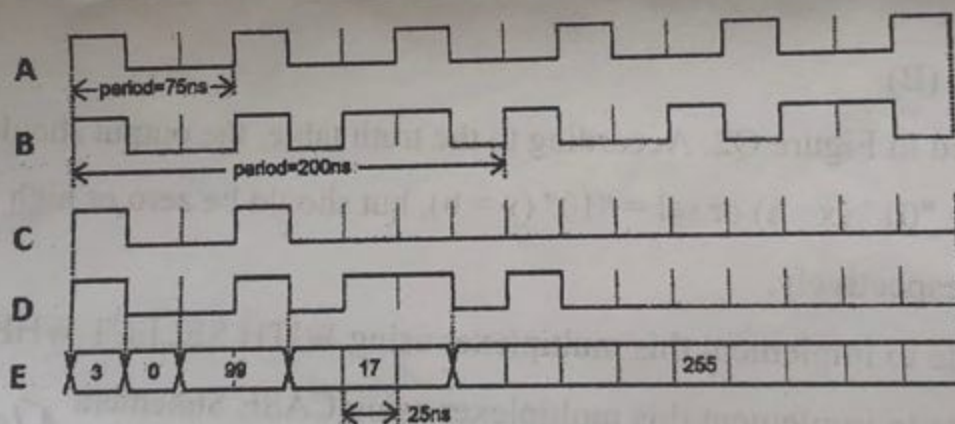


Figure Q4

Q5 - Consider the state diagram in Figure Q5

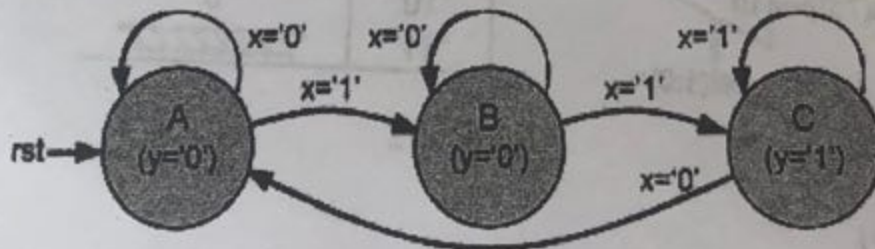


Figure Q5

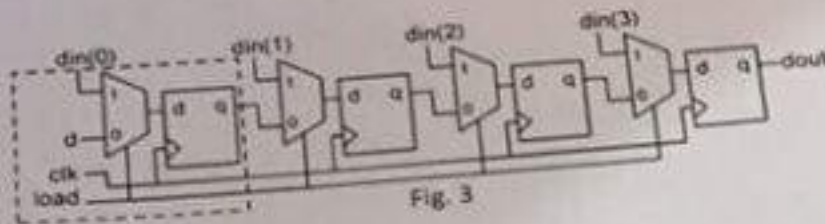


Fig. 3

Q3- Consider the state diagram in Figure 4

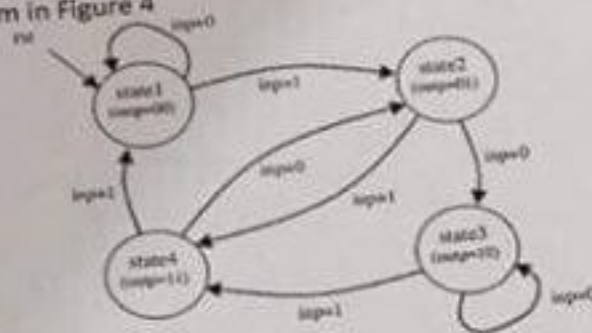


Fig. 4

- a) Draw the algorithmic state machine (ASM) of this state machine
 b) Complete the VHDL code of state transition rules and output generation for the state machines in the following template

```

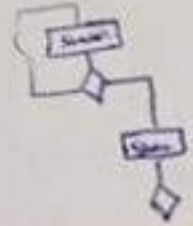
LIBRARY ieee;
USE ieee.stdlogic1164.all;
ENTITY fsm IS
PORT ( inp, clk, rst: IN STD_LOGIC;
      outp: OUT STD_LOGIC_VECTOR (1 DOWNTO 0));
END fsm;

ARCHITECTURE fsm OF fsm IS TYPE state IS (state1, state2, state3, state4);
SIGNAL present_state, next_state: state;
BEGIN
  PROCESS (rst, clk)
  BEGIN
    IF (rst='1') THEN
      present_state <= state1;
    ELSIF (clk'EVENT AND clk='1') THEN
      present_state <= next_state;
    END IF;
  END PROCESS;

  PROCESS (inp, present_state) BEGIN
    CASE present_state IS
      WHEN state1 =>
        -- State 1 implementation
      WHEN state2 =>
        -- State 2 implementation
      WHEN state3 =>
        -- State 3 implementation
      WHEN state4 =>
        -- State 4 implementation
    END CASE;
  END PROCESS;

  -- Moore output computation (outp)
END fsm;

```



END OF QUESTIONS

Q1/ Consider a circuit name DUT which has the following Entity definition:

```
ENTITY DUT IS
PORT (clk, rst, x, y: IN STD_LOGIC;
      Sig3: IN STD_LOGIC_VECTOR ( downto )
      Q: OUT STD_LOGIC_VECTOR ( downto ));
END ENTITY;
```

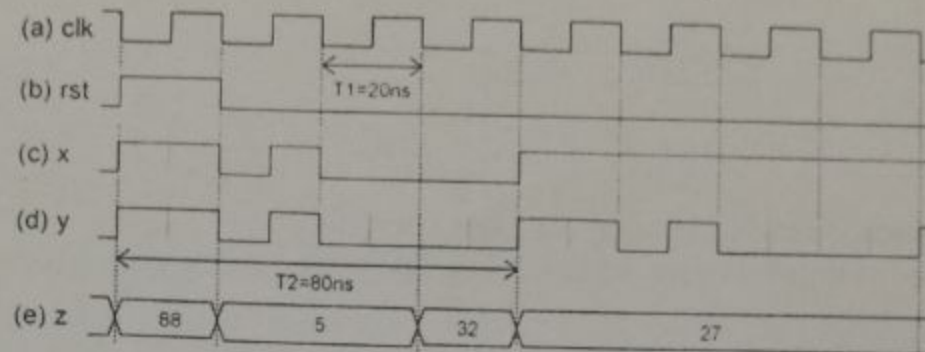


Figure Q1

Write a complete VHDL code of the processes that generates the signals clk, rst, x, y and z of figure Q1. Employ WAIT FOR statement. Each stimuli should be generated in a separate process. (10 Marks)

```
-----
LIBRARY ieee;
USE ieee.std_logic_1164.all;
-----
ENTITY testbench IS
END ENTITY;
-----
ARCHITECTURE testbench OF testbench IS
  constant clk_period : time := 20ns;
  SIGNAL clk: STD_LOGIC := '0';
  SIGNAL rst: STD_LOGIC := '0';
  -- define other signals here

  BEGIN
    --Generation of rst with WAIT FOR:

    clk PROCESS
    BEGIN
      clk <= '0';
      WAIT FOR clk_period/20ns;
      clk <= '1';
      WAIT FOR clk_period/20ns;
    END PROCESS;

    --Generation of rst with WAIT FOR:
    rst: PROCESS
    BEGIN
      --generation of x, y and z with WAIT FOR in separate processes

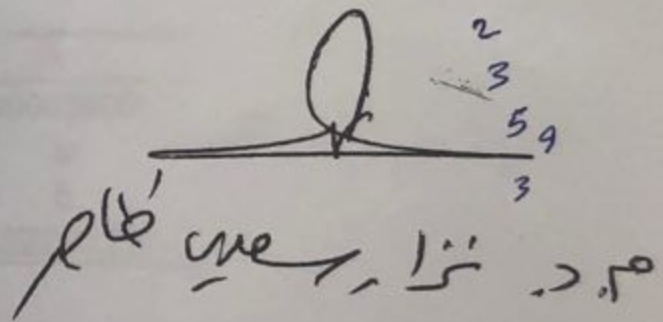
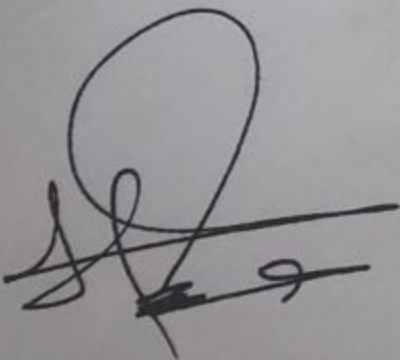
    END PROCESS;
  END ARCHITECTURE;
```

Q2- Consider the state diagram in Figure Q2

- A. Draw the algorithmic state machine (ASM) of this state machine (5 Marks)
B. Complete the VHDL code of state transition rules and output generation for the state machines in the following template (5 MARKS)

```
LIBRARY ieee;  
USE ieee.std.logic.1164.all;  
ENTITY fsm IS  
PORT ( X, clk, rst: IN STD_LOGIC;  
       Y: OUT STD_LOGIC);  
END fsm;  
  
ARCHITECTURE fsm_arch OF fsm IS  
TYPE state IS (A, B, C);  
SIGNAL present_state, next_state: state;  
  
BEGIN  
  PROCESS (rst, clk)  
  BEGIN  
    IF (rst='1') THEN  
      -- reset state  
    ELSIF (clk'EVENT AND clk='1') THEN  
      -- state transition  
    END IF;  
  END PROCESS;  
  
  PROCESS (x, present_state) BEGIN  
    CASE present_state IS  
      WHEN A =>  
        -- A implementation  
      WHEN B =>  
        -- B implementation  
      WHEN C =>  
        -- C implementation  
    END CASE;  
  END PROCESS;  
  
  -- Moore output computation (y)  
END fsm_arch;
```

-----END OF QUESTIONS, GOOD LUCK!



Handwritten calculations: $15 \times 3 = 45$

Q1/A1

1) a) "0111" NAND "0101" = "1010"

2) b) "0111" XOR "0010" = "0101"

c) "1001" NOR "1101" = "0010"

d) "101" XNOR "100" = "110"

e) "010" & "110" = "010110"

f) "11100"



Q1: Say that $a(7:0) = "00110011"$ and $b(3:0) = "1111"$. Determine the values produced by the assignments below. (answer only 5) (5 marks)

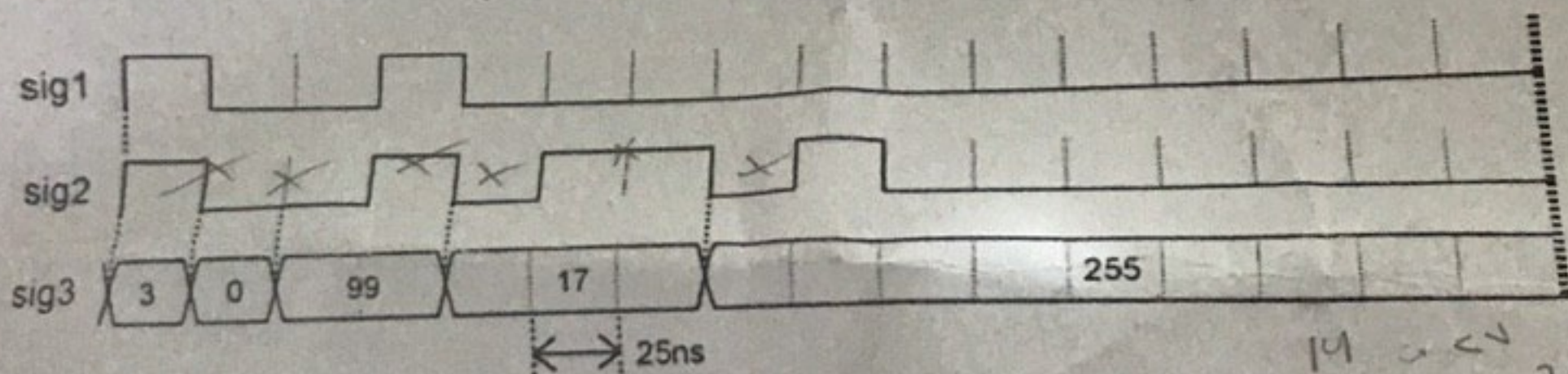
- a) $a(7 \text{ DOWNTO } 4) \text{ NAND } "0111"$
- b) $a(7 \text{ DOWNTO } 4) \text{ XOR NOT } b$
- c) $"1111" \text{ NOR } b$
- d) $b(2 \text{ DOWNTO } 0) \text{ XNOR } "101"$
- e) $a(2 \text{ DOWNTO } 0) \& "000"$
- f) $a(5) \& b(5) \& a(5 \text{ DOWNTO } 2)$

5432 1100

Q2: Implement a 4x1 multiplexer with: (10 marks)

- A- where inputs are single bit and using only logical operators (gate level design).
- B- N-bit inputs specify N using GENERIC and using WHEN statement.

Q3: Write the testbench code to generate the signals in the figure bellow: (10 marks)



Q4: Write the code for a zero to nine counter with reset signal input and the output (state value) is displayed using 7-segment displays HEX0. (15 marks)

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Good luck although you shouldn't need it!

Q4) (10 Marks)

- (a) Assuming architecture C in your lecture notes, determine the addressing mode and describe the following instructions as one of the following (a) Register-based, (b) Data transfer or (c) Control (ANSWER 5 ONLY)

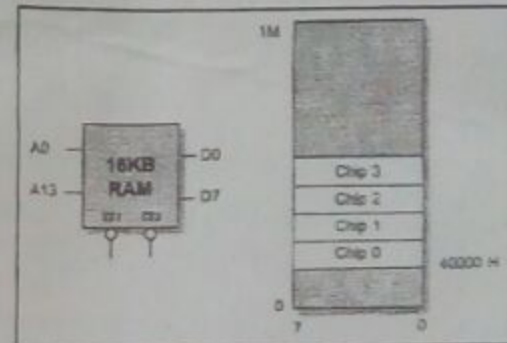
SUB R4,R2,R3 ; AND R1,R2,R3 ; ROR R1,R2,#3 ; STORO R1, R2,#1234 ; JMP #1234 ; BRZ R1,#1234

(5 Marks)

- (b) Assuming Architecture C in your lecture notes, write the instruction word for the instruction MOVI R3, #1234

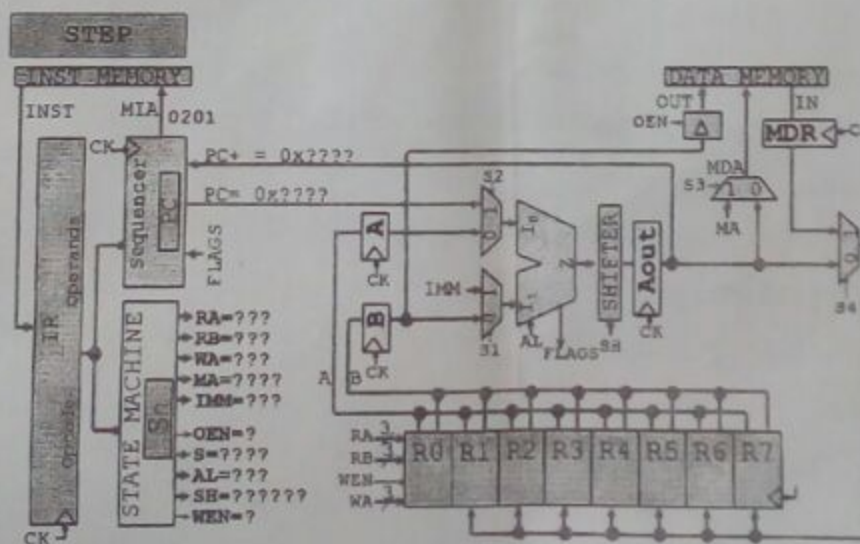
(5 Marks) (opcode for movi is 10000)

- (c) Show how to connect 4 of the following 16KB memory module starting from memory location 30000H onwards to produce 64KB memory capacity as shown in the memory map. Show the appropriate decoding logic for each module. Assume 20 bit address bus (A19-A0).



- Q5) Fill the following table with the control signals for each of the execution steps in architecture C (Multicycle) for the instruction BRNZ R3,0032 (10 Marks)

Step	PC=	PC+=	RA=	RB=	WA=	MA=	IMM=	OEN=	S[1-4]=	AL=	SH=	WEN=
	0x????	0x????	???	???	???	???	0x????	?	???	???	?????	?
Single Cycle	X	X										
Fetch	0032	0031								111		
Reg Read				011				0				0
Execute										101		
Mem Write	X	X	X	X	X	X	X	X	X	X	X	X
Reg Write	X	X	X	X	X	X	X	X	X	X	X	X



END OF QUESTIONS, GOOD LUCK!

- A. Design this circuit for 4 stages using VHDL (10 Marks)
- B. The entity shown in dashed box has the interfaces as follows:

```

ENTITY MUXED_FF IS
PORT (clk, D, S, Load : IN STD_LOGIC;
      Q : OUT STD_LOGIC);
END ENTITY;

```

Assume you have this entity in your library. Write VHDL to design the same circuit assuming the number of stages is a **GENERIC** named **N** and using **GENERATE** statement. (10 Marks)

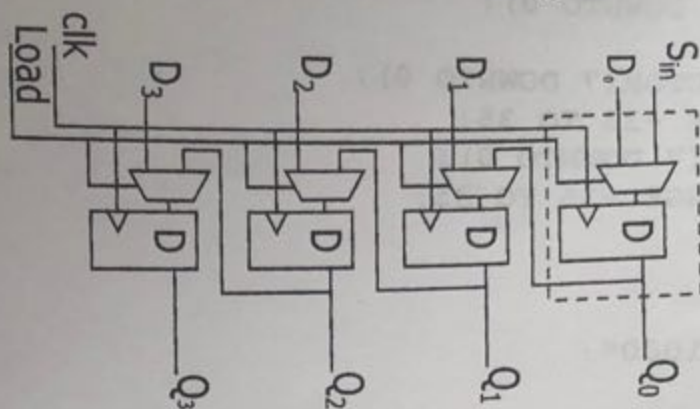


Figure Q3

Q4- Write a complete a testbench VHDL code of the processes that generate the signals A, B, C, D and E of figure Q4. Employ **WAIT FOR** statement. Each stimuli should be generated in a separate process. Note that A and B are periodic signals and should be generated indefinitely (10 Marks)

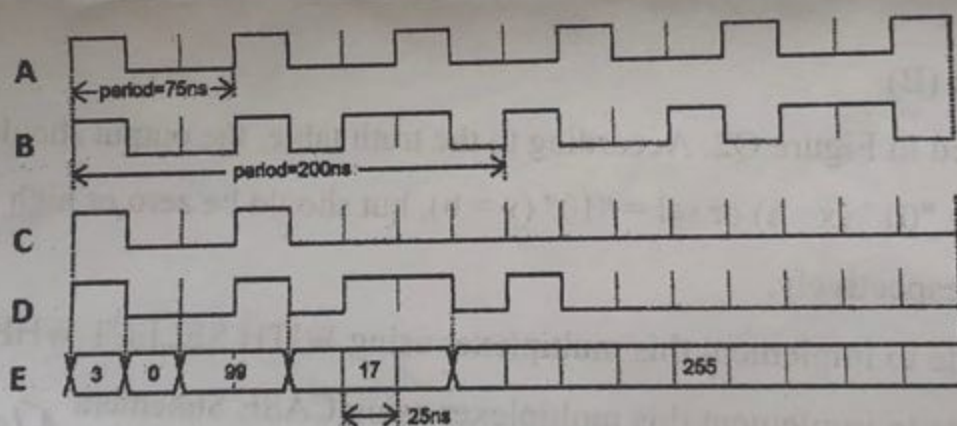


Figure Q4

Q5 - Consider the state diagram in Figure Q5

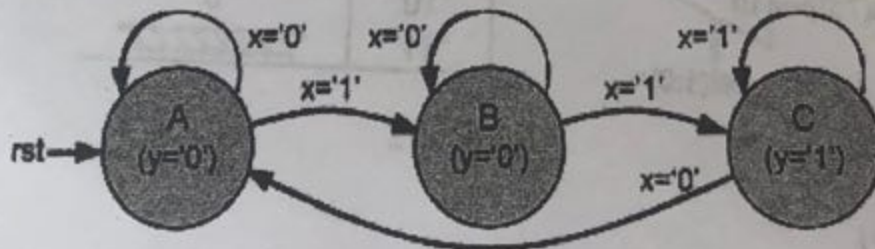


Figure Q5

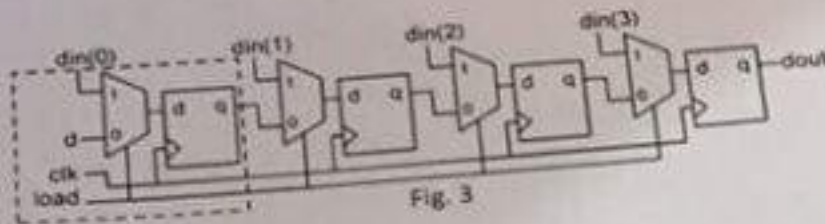


Fig. 3

Q3- Consider the state diagram in Figure 4

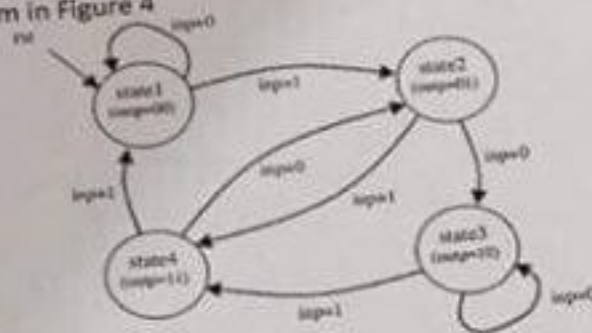


Fig. 4

- a) Draw the algorithmic state machine (ASM) of this state machine
 b) Complete the VHDL code of state transition rules and output generation for the state machines in the following template

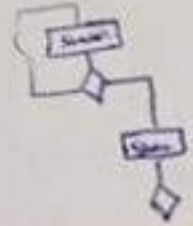
```

LIBRARY ieee;
USE ieee.stdlogic1164.all;
ENTITY fsm IS
PORT ( inp, clk, rst: IN STD_LOGIC;
      outp: OUT STD_LOGIC_VECTOR (1 DOWNTO 0));
END fsm;

ARCHITECTURE fsm OF fsm IS TYPE state IS (state1, state2, state3, state4);
SIGNAL present_state, next_state: state;
BEGIN
  PROCESS (rst, clk)
  BEGIN
    IF (rst='1') THEN
      present_state <= state1;
    ELSIF (clk'EVENT AND clk='1') THEN
      present_state <= next_state;
    END IF;
  END PROCESS;

  PROCESS (inp, present_state) BEGIN
    CASE present_state IS
      WHEN state1 =>
        -- State 1 implementation
      WHEN state2 =>
        -- State 2 implementation
      WHEN state3 =>
        -- State 3 implementation
      WHEN state4 =>
        -- State 4 implementation
    END CASE;
  END PROCESS;

  -- Moore output computation (outp)
END fsm;
  
```



END OF QUESTIONS

Q1/ Consider a circuit name DUT which has the following Entity definition:

```

ENTITY DUT IS
PORT (clk, rst, x, y: IN STD_LOGIC;
      Sig3: IN STD_LOGIC_VECTOR ( downto )
      Q: OUT STD_LOGIC_VECTOR ( downto ));
END ENTITY;

```

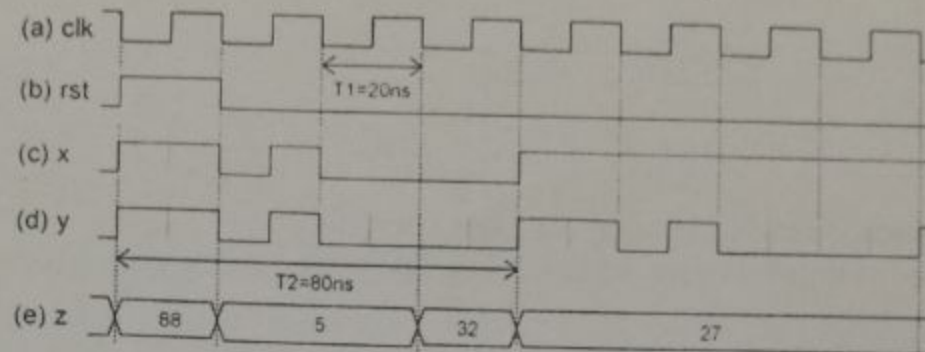


Figure Q1

Write a complete VHDL code of the processes that generates the signals clk, rst, x, y and z of figure Q1. Employ WAIT FOR statement. Each stimuli should be generated in a separate process. (10 Marks)

```

-----
LIBRARY ieee;
USE ieee.std_logic_1164.all;
-----
ENTITY testbench IS
END ENTITY;
-----
ARCHITECTURE testbench OF testbench IS
  constant clk_period : time := 20ns;
  SIGNAL clk: STD_LOGIC := '0';
  SIGNAL rst: STD_LOGIC := '0';
  -- define other signals here

  BEGIN
    --Generation of rst with WAIT FOR:

    clk PROCESS
    BEGIN
      clk <= '0';
      WAIT FOR clk_period/20ns;
      clk <= '1';
      WAIT FOR clk_period/20ns;
    END PROCESS;

    --Generation of rst with WAIT FOR:
    rst: PROCESS
    BEGIN
      --generation of x, y and z with WAIT FOR in separate processes

    END PROCESS;
  END ARCHITECTURE;

```

clk period

Q2- Consider the state diagram in Figure Q2

- A. Draw the algorithmic state machine (ASM) of this state machine (5 Marks)
- B. Complete the VHDL code of state transition rules and output generation for the state machines in the following template (5 MARKS)

```

LIBRARY ieee;
USE ieee.std.logic.1164.all;
ENTITY fsm IS
PORT ( X, clk, rst: IN STD_LOGIC;
      Y: OUT STD_LOGIC);
END fsm;

ARCHITECTURE fsm_arch OF fsm IS
TYPE state IS (A, B, C);
SIGNAL present_state, next_state: state;

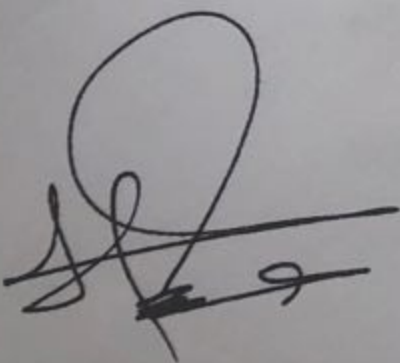
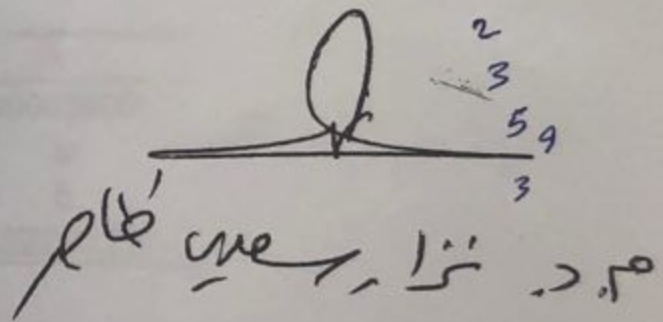
BEGIN
  PROCESS (rst, clk)
  BEGIN
    IF (rst='1') THEN
      -- reset state
    ELSIF (clk'EVENT AND clk='1') THEN
      -- state transition
    END IF;
  END PROCESS;

  PROCESS (x, present_state) BEGIN
    CASE present_state IS
      WHEN A =>
        -- A implementation
      WHEN B =>
        -- B implementation
      WHEN C =>
        -- C implementation
    END CASE;
  END PROCESS;

  -- Moore output computation (y)
END fsm_arch;

```

-----END OF QUESTIONS, GOOD LUCK!

Handwritten calculations: 2, 3, 5, 9, 3

Q1/A1

1) a) "0111" NAND "0101" = "1010"

2) b) "0111" XOR "0010" = "0101"

c) "1001" NOR "1101" = "0010"

d) "101" XNOR "100" = "110"

e) "010" & "110" = "010110"

f) "11100"



Note: Answer all questions

Q1- Fill the boxes with (✓) for the legal operation and (×) for illegal one, answer 10 only (10 Marks)

```
SIGNAL s1: BIT;
SIGNAL s2: BIT_VECTOR(7 DOWNT0 0);
SIGNAL s3: STD_LOGIC;
SIGNAL s4: STD_LOGIC_VECTOR(7 DOWNT0 0);
SIGNAL s5: INTEGER RANGE -35 TO 35;
VARIABLE v1: BIT_VECTOR(7 DOWNT0 0);
VARIABLE v2: INTEGER RANGE -35 TO 35;
```

- ✓ 1. s2(7) <= s1;
- × 2. s3 <= s4(0);
- ✓ 3. s2 <= v1 XOR "10001000";
- ✓ 4. s5 <= v2/2;
- ✓ 5. s3 <= 'Z';
6. s2 <= (OTHERS=>'0');
7. v2 := 35;
- ✓ 8. v1 := "11110000";
- × 9. s1(0) <= s2(0);
- ✓ 10. s3 <= s1 OR s2(2);
- × 11. s2 <= (8=>'0', OTHERS=>'Z');
- ✓ 12. v2 <= -35;

Q2- ANSWER (A) OR (B)

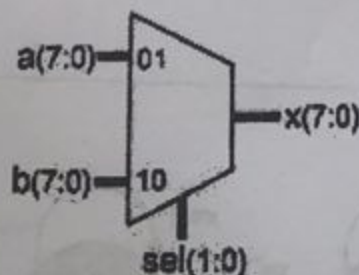
A multiplexer is depicted in Figure Q2. According to the truth table, the output should be equal to one of the inputs if sel = "01" (x = a) or sel = "10" (x = b), but should be zero or high impedance if sel = "00" or sel = "11", respectively.

A- Write VHDL code to implement this multiplexer using WITH SELECT WHEN Statement

B- Write VHDL code to implement this multiplexer using CASE Statement

[10 MARKS]

[10 MARKS]



sel	x
"00"	"00000000"
"01"	a
"10"	b
"11"	"ZZZZZZZZ"

Figure Q2

Q3- ANSWER (A) OR (B)

Consider the parallel adder shown in Figure Q3. Assuming you have this entity in your project do the following:

سؤال 1

```
;signal Y : integer
```

:Select one

- ☐ ;a. X <=to_integer(unsigned(Y))
- ☐ ;b. Y <=to_integer(unsigned(X))
- ☐ ;c. X <=to_integer(Y)
- ☐ ;d. Y <=to_integer(X)

[CLEAR MY CHOICE](#)

التالي <

Question 1

Not yet answered

Marked out of 1.00

Flag question

The truth table below represents a FSM with a 1-bit input (x) and a 1-bit output y . Suppose the machine is initialized to its starting state '00' and then processes the input values: 1,0,0,1,0,0 (1 input bit after each Clk). What are the final state?

Present State		Next State				Output	
		$x = 0$		$x = 1$		$x = 0$	$x = 1$
A	B	A	B	A	B	y	y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0

Select one:

- ☐ a. 10
- ☐ b. 01
- ☐ c. 11
- ☐ d. 00

Quiz navigation

1 2 3

7 8 9

Finish attempt ...

Time left 0:49:11

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2



ieee.std_logic_1164.all :

ENTITY part4_vhd_tst IS

END part4_vhd_tst :

العمارة part4_arch OF part4_vhd_tst IS

Clock_Cycle: time: = 40ns:

SIGNAL Clk: STD_LOGIC :

SIGNAL SCLR: STD_LOGIC :

SIGNAL D: STD_LOGIC ,

SIGNAL O: STD_LOGIC

عنصر 4 تسجيل

PORT (

Clk D, SCLR: IN STD_LOGIC:

O: OUT STD_LOGIC

) :

عنصر النهاية :

ابدأ

i1: 4SRegister

PORT MAP (

سرد الاتصالات بين المنافذ الرئيسية -

والإشارات

Clk => Clk ,

D => D ,

SCLR => SCLR ,

O => O





2



Question 1

Not yet answered

Marked out of 2.00

Flag question

Given the Testbench code bellow that aim to test a 8-bit shift register with serial input D, Clear input (SCLR operate on active high), and a clock (CLK). This shift register operate on +ev edge. Then what are the content of this register would be after 8 clock cycles with shift left:

```
LIBRARY ieee;  
USE ieee.std_logic_1164.all;  
  
ENTITY part4_vhd_tst IS  
END part4_vhd_tst;  
  
ARCHITECTURE part4_arch OF part4_vhd_tst  
IS
```



الإنجليزية



Question 3

Not yet answered

Marked out of
1.00

Flag question

Given the the signals bellow:

SIGNAL s1: BIT;
SIGNAL s2: BIT_VECTOR(7 DOWNT0 0);
SIGNAL s3: STD_LOGIC;
SIGNAL s4: STD_LOGIC_VECTOR(7 DOWNT0 0);
VARIABLE v1: BIT_VECTOR(7 DOWNT0 0);

Select which statement bellow are legal (more than one choice are allowed)

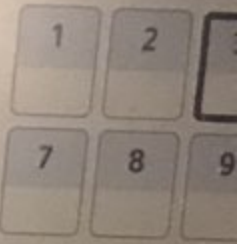
Select one or more:

- ☐ s2 (5 DOWNT0 0); <= S2 (6 DOWNT0 2);;
- ☐ s2 <= 'Z';
- ☐ s2 <= "10101011";
- ☐ s2(3) <= s1;

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Quiz naviga



Finish attempt ..

Time left 0:36:37

PREVIOUS ACTIVITY

Report submission Lab Part II



2



Question 3

Not yet answered

Marked out of 1.00

Flag question

Given the signals bellow

signal X : std_logic_vector(7 downto 0);

signal Y : unsigned(7 downto 0);

Which of the following is a legal VHDL assignment

Select one:

- ☐ a. X <= unsigned(Y);
- ☐ b. X <= std_logic_vector(Y);
- ☐ c. Y <= std_logic_vector(X);
- ☐ d. Y <= to_integer(X);





2



BEGIN

-- regn (R, Clock, Resetn, Q)

A_reg: regn PORT MAP (SW, KEY(1), KEY(0),
A);

B <= SW;

-- drive the displays through 7-seg decoders

digit_3: hex7seg PORT MAP (A



, HEX3);

digit_2: hex7seg PORT MAP (A



, HEX2);

digit_1: hex7seg PORT MAP (B



, HEX1);

digit_0: hex7seg PORT MAP (B



, HEX0);

S <=



;

carry <= S(8);

digit_5: hex7seg PORT MAP (S



Question 3

Not yet answered

Marked out of
1.00

Flag question

Given the the signals bellow:

SIGNAL s1: BIT;
SIGNAL s2: BIT_VECTOR(7 DOWNT0 0);
SIGNAL s3: STD_LOGIC;
SIGNAL s4: STD_LOGIC_VECTOR(7 DOWNT0 0);
VARIABLE v1: BIT_VECTOR(7 DOWNT0 0);

Select which statement bellow are legal (more than one choice are allowed)

Select one or more:

- ☐ s2 (5 DOWNT0 0);<= S2 (6 DOWNT0 2);
- ☐ s2 <= 'Z';
- ☒ s2 <= "10101011";
- ☒ s2(3) <= s1;

Quiz navigation

1	2	3
7	8	9

Finish attempt ...

Time left 0:31:56

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PREVIOUS ACTIVITY

Report submission Lab 7 Part II

Question 2

Not yet answered

Marked out of
1.00

Flag question

Given the the signals bellow:

SIGNAL s1: BIT;

SIGNAL s2: BIT_VECTOR(7 DOWNT0 0);

SIGNAL s3: STD_LOGIC;

SIGNAL s4: STD_LOGIC_VECTOR(7 DOWNT0 0);

VARIABLE v1: BIT_VECTOR(7 DOWNT0 0);

Select which statement bellow are legal (more than one choice are allowed)

Select one or more:

☐ s2 (5 DOWNT0 0) <= s2 (7 DOWNT0 2);

☐ s1 <= '11';

☐ s2(4) <= 'Z';

☐ s2(3) <= s1;